UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 002 · Nernst equation and its derivation · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·4	Deck 002, slide 4 (rendered .pptx)
T001 00:21:42	Class 001 AV-summary transcript (covers classes 1 and 2), segment start
PDF p.13	Consolidated Unit 2 slide PDF (older syllabus), page 13
ESA Jan-25 s1 2a	Past paper: Jan 2025, set 1, question 2.a

Contents

1. Part A · Formula sheet
2. 1 · Free energy and electrode potential
3. 2 · Derivation of the Nernst equation for a single electrode
4. 3 · Using the single-electrode equation: concentration and temperature
5. 4 · Nernst equation for a cell
6. Part B · Symbol glossary
7. Part C · Notation and conversions

A Formula sheet

1 Free energy and electrode potential

E1.1 · Maximum work and free energy

ESA ★★

$$-\Delta G = W_{max}$$

GIVES	Maximum useful work from a reversible process.
APPLIES	Constant T and P, reversible operation.
UNITS	J (or J mol^{-1})
NOTE	Decrease in free energy = maximum work.

S002·3 · T001 00:19:38

E1.2 · Maximum electrical work

ESA ★★★

$$W_{max} = \underbrace{nF}_{\text{charge}} \times \underbrace{E}_{\text{energy/charge}} = nFE$$

GIVES Electrical work from n mol of electrons at potential E.

APPLIES Any electrode or cell.

UNITS C mol^{-1} × V = J mol^{-1}

NOTE 1 V = 1 J C^{-1}.

S002·3 · T001 00:20:21 · T001 00:21:06

E1.3 · Free energy and potential

ESA ★★★

$$\Delta G = -nFE, \qquad \Delta G^\circ = -nFE^\circ$$

GIVES Links thermodynamics to potential.

APPLIES Second form: standard conditions (1 M, 1 atm, 298 K).

UNITS J mol^{-1}

NOTE $E > 0 \Rightarrow \Delta G < 0$: spontaneous.

S002·3 · T001 00:16:25 · T001 00:21:24

2 Derivation of the Nernst equation for a single electrode

E2.1 · Free energy and reaction quotient

ESA ★★★

$$\Delta G = \Delta G^\circ + RT \ln Q$$

GIVES How ΔG changes as Q changes.

APPLIES Any reaction; ln is the natural log.

UNITS J mol^{-1}

NOTE No nF here: $RT \ln Q$, not $\frac{RT}{nF} \ln Q$.

S002·4 · T001 00:21:42

E2.2 · Reaction quotient for the electrode

ESA ★★★

$$\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}, \qquad Q = \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$$

GIVES Q for the reduction.

APPLIES Electrode written as a reduction.

UNITS dimensionless

NOTE Product over reactant.

S002·4 · T001 00:22:05

E2.3 · After substituting and dividing by −nF

ESA ★★★

$$-nFE = -nFE^\circ + RT\ln\frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]} \implies E = E^\circ - \frac{RT}{nF}\ln\frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$$

GIVES The general Nernst equation, before simplification.

APPLIES Any T.

UNITS V

NOTE Dividing by $-nF$ flips the sign of the RT term.

S002·4 · T001 00:22:32 · T001 00:23:01

E2.4 · Single electrode, natural log

ESA ★★★

$$E = E^\circ + \frac{RT}{nF}\ln[\mathrm{M^{n+}}]$$

GIVES Electrode (reduction) potential at any T.

APPLIES $[\mathrm{M}] = 1$ for the pure metal.

UNITS V

NOTE $-\ln\frac{1}{x} = +\ln x$.

S002·5 · T001 00:23:17

E2.5 · Single electrode at 298 K

ESA ★★★

$$E = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$$

GIVES The form used in numericals.

APPLIES 298 K; reduction potential.

UNITS V

NOTE $0.0591 = 2.303 \times \frac{8.314\times298}{96500}$.

S002·5 · T001 00:23:29 · T001 00:23:50

3 Using the single-electrode equation: concentration and temperature

E3.1 · The Nernst factor

ESA ★★

$$\frac{2.303\,RT}{F} = \frac{2.303\times8.314\times298}{96500} = 0.0591\ \mathrm{V}$$

GIVES The 298 K constant.

APPLIES Slide constants; sympy gives 0.05913.

UNITS V

NOTE CODATA constants at 298.15 K give 0.05916 V; both round to 0.0591.

S002·5 · T001 00:23:40

E3.2 · Any temperature

ESA ★

$$E = E^\circ + \frac{2.303\,RT}{nF}\log[\mathrm{M^{n+}}]$$

GIVES Electrode potential at temperature T.

APPLIES $T \neq 298$ K: recompute the factor (0.0611 V at 308 K).

UNITS V

NOTE Assumes E° is the 298 K value unless given.

ADDED FOR CLARITY S002·5 · T001 00:19:25

E3.3 · Change per decade

ESA ★

$$\Delta E = \frac{0.0591}{n} \text{ per tenfold change in } [\mathrm{M^{n+}}]$$

GIVES Slope of E vs $\log[\mathrm{M^{n+}}]$.

APPLIES 298 K.

UNITS V

NOTE $n = 1$: 59.1 mV; $n = 2$: 29.6 mV; $n = 3$: 19.7 mV.

ADDED FOR CLARITY S002·5

4 Nernst equation for a cell

E4.1 · Nernst equation for a cell (298 K)

ESA ★★★

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

GIVES Cell emf at non-standard concentrations.

APPLIES $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$; n from the balanced equation.

UNITS V

NOTE General T: replace 0.0591 by $2.303RT/F$.

S002·5 · T001 00:24:15 · T001 00:25:06

E4.2 · Reaction quotient for a cell

ESA ★★★

$$Q = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

GIVES Products over reactants with coefficients as powers.

APPLIES Pure solids and liquids have activity 1 and are left out.

UNITS dimensionless

NOTE Al–Cu cell: $Q = [\mathrm{Al}^{3+}]^2/[\mathrm{Cu}^{2+}]^3$.

S002·5 · T001 00:24:15

E4.3 · Cell emf from electrode potentials

ESA ★★★

$$E_{cell} = E_{right} - E_{left} = E_{cathode} - E_{anode}$$

GIVES Cell emf from two reduction potentials.

APPLIES Both from E2.5 at their own concentrations.

UNITS V

NOTE Gives the same answer as E4.1.

T001 00:16:00 · PDF p.10

E4.4 · Equilibrium constant

ESA ★

$$E_{cell} = 0,\ Q = K \implies \log K = \frac{nE^\circ_{cell}}{0.0591}, \qquad \Delta G^\circ = -2.303\,RT\log K$$

GIVES K from E°_{cell}.

APPLIES At equilibrium (the cell is “dead”).

UNITS K dimensionless

NOTE Not taught in class 002; a short note only.

ADDED FOR CLARITY NOT ON CURRENT SLIDES S002·5

B Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
E	Electrode potential / cell emf	Reduction potential in E2.5; cell emf in E4.1 (same letter on the slides)	V	E1.2–E4.1
E°	Standard electrode potential	At 1 M, 1 atm, 298 K	V	E1.3, E2.3–E2.5
E°_{cell}	Standard emf	$E^\circ_{cathode} - E^\circ_{anode}$	V	E4.1, E4.4
ΔG, ΔG°	Free-energy change	Actual / standard	J mol^{-1}	E1.1, E1.3, E2.1
W_{max}	Maximum work	$= -\Delta G = nFE$	J	E1.1, E1.2
n	Electrons transferred	Moles of electrons in the balanced reaction	—	all
F	Faraday constant	96500 C mol^{-1}	C mol^{-1}	E1.2, E2.3
R	Gas constant	8.314 J K^{-1} mol^{-1}	J K^{-1} mol^{-1}	E2.1, E3.1
T	Temperature	In kelvin	K	E2.1, E3.2
Q	Reaction quotient	Products/reactants at any instant	—	E2.1, E2.2, E4.2
K	Equilibrium constant	Q at equilibrium	—	E4.4
[M^{n+}]	Ion concentration	Molar concentration of metal ions	mol L^{-1}	E2.2–E3.2
[M]	Metal	Pure solid: taken as 1	—	E2.2, E2.3
ln, log	Logarithms	Base e / base 10; $\ln x = 2.303 \log x$	—	E2.4, E2.5

Remark. E means the electrode potential in the single-electrode equation (E2.4) and the cell emf in the cell equation (E4.1); the slides use the same letter for both. n is the number of electrons in the *balanced* reaction, which is 6 (not 2 or 3) for the Al–Cu cell. log is base 10 and ln base e.

C Constants and conversions used in this class

Quantity	Value used on the slides	Remark
R	$8.314\ \mathrm{J\,K^{-1}\,mol^{-1}}$	gas constant
F	$96500\ \mathrm{C\,mol^{-1}}$	charge on one mole of electrons (exact 96485)
T	$298\ \mathrm{K}$	25 °C; standard temperature
$\ln x$	$2.303 \log_{10} x$	$\ln 10 = 2.3026$
RT/F **at 298 K**	0.02567 V	sympy
$2.303RT/F$ **at 298 K**	0.0591 V	0.05913 with slide constants; 0.05916 with CODATA at 298.15 K
$2.303RT/F$ **at 308 K**	0.0611 V	scales with T
1 V	$1\ \mathrm{J\,C^{-1}}$	so nFE is in J